

Building a Metaverse for All: Opportunities and Challenges for Future Inclusive and Accessible Virtual Environments

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ABSTRACT

The Metaverse is proposed as a collection of interconnected virtual worlds allowing people to seamlessly access and traverse through them. It offers an opportunity to remove physical boundaries and borders present in our reality with the potential to bring together people from all walks of life. The development of a Metaverse or a “new world” offers a compelling opportunity to establish a space that is both all-encompassing and accessible to everyone. However a key challenge is understanding how the Metaverse can be designed from the ground up to be inclusive and accessible. This workshop aims to gain further understanding of how to create an open and inclusive Metaverse for all, while also exploring methods for its evaluation. The key outcomes of this workshop outline new opportunities for improving the inclusivity of the Metaverse, evaluation methodologies, and key considerations for designing accessible Metaverse environments and interactions.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Accessibility; Virtual reality; Mixed / augmented reality.**

KEYWORDS

Human Computer Interaction, Metaverse, Inclusivity, Accessibility, Virtual Reality, Augmented Reality

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1 BACKGROUND

In recent years, virtual reality (VR) has evolved into a widely adopted technology, finding extensive use in various educational, research and commercial applications. It has been implemented for a diverse range of purposes, including video games, simulations, training, and facilitating social interactions. These developments collectively point towards the emergence of the Metaverse, a long-standing concept describing a network of interconnected virtual worlds, each governed by distinct entities. While it shares similarities with the internet, the Metaverse is primarily accessed with VR headsets, providing users with immersive experiences in full 3D environments [4, 17]. Envisioned as a persistent parallel world [1], the Metaverse is set to utilise technologies beyond VR, leveraging technologies like Blockchain to establish a digital economy and enable asset ownership [3, 11], and Artificial Intelligence to automate building digital assets and drive realistic non-human virtual actors in virtual environments [12].

Companies like Meta and Apple are actively developing platforms for their respective VR devices. While there are shared characteristics, the actual functionalities of these products can significantly differ. This divergence has the potential to fragment design approaches for the Metaverse due to “walled gardens” or centralisation by one company [5], contradicting the core philosophy of the Metaverse which advocates for a decentralised and open platform [14, 16].

Despite the potential of the Metaverse becoming pervasive in our daily lives and the race by major companies to establish it, there is a risk that part of the population will be excluded from the affordances of its design. This includes individuals living with disabilities, senior citizens, disadvantaged youth and women, LGBTQIA+ individuals,

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refugees and migrants, and First Nations people. Considering the diversity of our population is a critical step towards the Metaverse becoming a truly open and inclusive platform. The Metaverse and the social virtual environments within it could help connect these different groups with their family, friends, and teachers. Therefore, it is critical that the Metaverse becomes a truly open and inclusive platform that empowers humans, in a similar vein to the internet. To work towards this, there is an opportunity to leverage the extensive insights gained from over 60 years of VR research within HCI to enhance the core principles of the Metaverse and its virtual environments, particularly in the domains of inclusivity and accessibility.

Research that investigated accessible VR experiences explored alternative input mechanisms, such as mapping a cane for people with vision impairments to a VR environment [23]. Work by Wedoff et al. [21] demonstrated the potential of using sound in a VR game for people with vision impairments. Other work has presented a toolkit for developers to enhance VR environments for people with low vision [24]. Virtual environments could also leverage physiological signals such as breathing [18] and heart-rate [22] to enable natural interactions and reactions. Along with this, depth sensing cameras have enabled VR experiences to be adapted to people, such as people who use wheelchairs [8]. Accessibility features are also gradually being introduced to VR platforms. For instance, research by Teófilo et al. [20] highlighted that the Samsung Gear VR had the following accessibility features: Zoom, Inverted Colours, Screen Reader, and Captions. Work by Herskovitz et al. [10] has also explored accessibility of augmented reality (AR) apps and derived insights into improving their accessibility for people with vision impairments.

Alongside hardware and interaction design, it is important to comprehend the social dimensions of the Metaverse in order to empower individual users of diverse identities. For instance, community groups are currently leveraging virtual environments, as exemplified by the potential of the LGBTQIA+ community using virtual spaces to provide support for one another [7] and seek embodied visibility online [6, 13]. Future virtual spaces should be conceived with the aim of empowering communities, fostering safe environments where people can convene and freely express themselves.

Despite the growing research in this space, according to work by Heilemann et al. [9], there has been a lack of standard accessibility guidelines for VR developers to follow, along with a limited number of tools to help avoid common accessibility problems. This situation is improving however with Meta publishing their own guidelines on Designing Accessible VR¹ for their Meta Quest VR platform. Apple has also released some initial design guidelines for their VisionPro Headset². This covers aspects such as UX/UI, interactions, movement, display, audio, ergonomics, and accessibility.

Utilising this growing knowledge of accessible immersive technology there is a need for researchers to come together and discuss how this knowledge can be utilised for a Metaverse and what further knowledge is needed to achieve one which is inclusive for

users with various accessibility needs. Alongside considerations for accessibility and possible standards for design, there is also a need to understand how the accessibility of an immersive Metaverse environment can be evaluated.

As our understanding of immersive technology continues to expand, it becomes imperative for researchers to collaborate and explore how existing knowledge can be harnessed and what additional new knowledge is needed to create a truly inclusive and accessible Metaverse for people with diverse needs. In addition to addressing inclusivity and accessibility, it is equally essential to understand best practices for evaluating how effectively virtual environments within the Metaverse accommodate these considerations.

2 WORKSHOP THEMES AND TOPICS

This workshop builds on our successful CHI 2023 workshop, “Towards an Inclusive and Accessible Metaverse” [15]. Like the previous workshop, we aim to advance the HCI community’s knowledge on how Metaverse platforms and the virtual environments within them can be designed to foster inclusivity and accessibility. In this workshop we also aim to gain more understanding on how the design process can be improved, enabling future Metaverse platforms to get it right from the start. Attendees can share aspects such as:

2.1 Theme 1: Conceptualisation and prototyping

Understanding of methods for prototyping an inclusive and accessible Metaverse, such as virtual environments, interfaces, and interactions. Topics can include:

- Prototyping techniques (low-fi through to hi-fi).
- Tools and platforms to support prototyping.
- Potential design considerations for metaverse environments and interactions.

2.2 Theme 2: Methods and metrics for evaluating inclusivity and accessibility

Identifying strategies and best practices to evaluate the accessibility of existing and future Metaverse experiences. We are interested in compiling effective research methodologies and approaches which can be leveraged by not only researchers, but also designers and developers in the industry. Topics can include:

- Understanding how to improve the inclusiveness of virtual environments.
- Potential new methods and metrics for evaluating accessibility issues with virtual environments.
- New methodologies and approaches to running studies.

2.3 Theme 3: Future considerations for the Metaverse

This theme relates to broader discussions around the current state of the Metaverse, what the Metaverse should be in the future, and what challenges need to be overcome to achieve the desired state. This can include topics such as:

¹Designing Accessible VR - <https://developer.oculus.com/resources/design-accessible-vr/>

²Designing for visionOS - <https://developer.apple.com/design/human-interface-guidelines/designing-for-visionos/>

- Current and future accessibility challenges facing the metaverse. What does an accessible metaverse and the environments/worlds within it look like?
- Discussion of ability-based design as it relates to VR and the Metaverse.
- Work towards the establishment of design guidelines and speculation on how they can be enforced within a decentralised Metaverse platform. Whose responsibility is it?

3 PRE-WORKSHOP PLANS

3.1 Target Audience

The overarching goal of this workshop is to bring together researchers and practitioners from various disciplines and backgrounds, to jointly develop a research agenda, to understand the challenges and opportunities for the Metaverse from the lens of inclusivity and accessibility for all possible stakeholders, including people living with disabilities, senior citizens, and children. Therefore, we aim to broadly advertise the workshop to different communities of researchers and practitioners. We will select participants based on the potential contribution they can offer to the workshop agenda.

3.2 Recruitment

We will distribute the call for participation through posting announcements to lists such as CHI-ANNOUNCEMENTS and social media platforms, namely LinkedIn and Facebook. We will send targeted email invitations to researchers in our own networks and to leading researchers across different academic institutions inviting them to participate and distribute the announcement within their organisations. Our website hosts the call for participation, information about the workshop's organisers, news and announcements, and paper submission instructions.

3.3 Paper Submission and Review Procedure

Submissions to this workshop will take the form of position papers (maximum 6 pages in single-column CEUR-WS Format, excluding references) addressing one of the themes of the workshop (cf. Section 2 Workshop Themes and Topics). Participants can also choose to submit a motivation statement (in the same format), describing their reasons for joining the workshop and/or what they would contribute towards the workshop agenda. They should also include a statement on the potential goals of their research and the problems it aims to address. Ultimately, the length is based on the weight of the contribution. Shorter, more focused papers are highly encouraged.

Submissions will be made through EasyChair. Following submission, the position papers will be divided for review among the workshop organisers and invited reviewers. Reviews will be based on quality and relevance to the themes of the workshop. After discussion of all submissions, successful submissions will be invited to the workshop. At this point participants will be asked to express any accessibility concerns which might affect their participation so that we can accommodate them accordingly.

Beyond the quality and relevance of submissions, we will aim to ensure an interdisciplinary and balanced group of researchers in this field. We will solicit widely and internationally for contributions to the workshop. This will both support the interactivity of the

networking activities and also reflect the growing relevance and potential of interdisciplinary research across a range of HCI sub-disciplines.

3.4 Workshop Format

This workshop will be held in-person between May 11 and 16 during CHI'24. To support participants that are unable to access in-person or synchronous virtual space, particularly for reasons relating to health, travel or technical limitations, we will be offering asynchronous materials that will be provided to participants to engage with. To help participants familiarise themselves with each other's work and interests ahead of the workshop, participants will prepare a short introduction to upload to our workshop website two weeks before the workshop. Links to these will be shared among all participants, alongside the submitted position papers. We will use the Miro platform to engage in interaction between both in-person participants and online remote during the workshop activities. We will create a Slack group for the workshop, allowing participants to discuss each others' papers asynchronously, ask questions and self-select discussion groups for activities at the workshop. We will provide a system for this group-selection (to be determined). As such, our one-day workshop will focus on meetings and interactions between the participants while also allowing researchers a brief time to present their own work.

4 WORKSHOP STRUCTURE

The workshop will run as a full-day in-person workshop, enabling us to bring people together face-to-face while also welcoming international participants synchronously via Zoom, using an organiser's institutional account, and will make use of captioning for hard-of-hearing users. During the workshop, participants with position papers will be asked to briefly present their work for 5 minutes followed by 2 minutes Q&A. Along with this, we will contribute to a larger discussion and workshop activities. This will be based around an expanded set of key themes identified in our participants' responses to the online registration questionnaire and from their position paper submissions. A key component of the workshop will be a speculative design [2, 19] challenge to engage workshop participants in envisioning scenarios set in the future, that allow participants to speculate on possible designs, experiences or interactions with Metaverse experiences which allow for greater accessibility. The workshop will foster new ideas and a build a sense of empathy among its participants.

Table 1 below outlines a preliminary schedule - to be taken as an example, and subject to change.

5 POST-WORKSHOP PLANS

The workshop proceedings will be published in CEUR-WS³. In addition, the discussions and findings from the workshop will be refined into a "manifesto" on the challenges and opportunities for increasing the accessibility of the Metaverse. This manifesto will provide the basis for a journal special issue, where participants with an excellent contribution to the workshop will be invited to submit an extended version of their position paper.

³<https://ceur-ws.org>

Table 1: Workshop schedule

Workshop Activity	Time (HST)
Welcome	9:00am
Keynote speaker	9:10am
Morning tea	9:50am
Presentation of position papers	10:00am
Lunch	12:00pm
Group activities	1:00pm
Afternoon tea	2:30pm
Group presentations	2:40pm
Workshop wrap-up	3:30pm

With the organisers' strong links in the relevant research communities it can be expected that beyond concrete plans the workshop contributes to further follow-up activities such as iterations of this workshop in future conferences. To facilitate such activities the workshop's website will continue to be regularly updated. Furthermore, we plan to foster a community of researchers focused on improving the accessibility of immersive technologies and the Metaverse through a Slack channel created for this workshop. This group will be maintained after the event to allow future collaborations and sharing of datasets, code, and best practices.

6 CALL FOR PARTICIPATION

The Metaverse has the potential to bring the world much closer together through its network of connected virtual worlds and decentralised nature. Considering its potential to reach global appeal it is important to ensure that the Metaverse is accessible for people with physical and intellectual disabilities. Building on our successful workshop at CHI'23, "Towards an Inclusive and Accessible Metaverse" [15], this workshop intends to work towards a broad understanding of how the Metaverse can be designed in order to achieve this, along with understanding how the success of this can be evaluated. To achieve this, our workshop focuses on three core themes: (1) conceptualisation and prototyping; (2) methods and metrics for evaluating inclusivity and accessibility; and (3) future considerations for the Metaverse.

We welcome submissions from interested researchers and practitioners for position papers (maximum 6 pages in single-column CEUR-WS Format, excluding references) which address at least one of the key theme(s) of the workshop and present original material. The paper should also include a statement on the potential goals of the research and the problem(s) it aims to address. Submissions will be made via EasyChair. For more information visit our website (<https://sites.google.com/view/accessiblemetaverse>) or contact callum.parker@sydney.edu.au.

Papers will be peer-reviewed by an international program committee and selected based on relevance to the workshop themes and quality of the work. At least one author of each accepted position paper must attend the workshop, and all participants must register for both the workshop and for at least one day of the conference.

7 ORGANISERS

The workshop has a broad international group of organisers, including established researchers, and younger perspectives; with interests in HCI, accessibility, metaverse.

Callum Parker is a Lecturer in Interaction Design at the Urban Interfaces Lab within the University of Sydney's School of Architecture, Design, and Planning. Callum's research seeks to gain new understanding of how interactive digital city interfaces can work for everyone, of all shapes and sizes, and abilities. His work is also dedicated to exploring the evolving concept of "public space" in the Metaverse and understanding how diverse people can engage with it. Callum utilises cutting-edge technologies, including pervasive displays and augmented, virtual, and mixed realities.

Soojeong Yoo is a Research Fellow at the UCL Interaction Centre and a member of the Wellcome/EPSRC Centre for Interventional and Surgical Sciences (WEISS). Her research interests include augmenting workplaces with novel technologies such as virtual reality (VR) and augmented reality (AR) within the context of physical activity, on-body interaction, personalised dashboards and human adapted HCI.

Joel Fredericks is a Lecturer in Design at The University of Sydney's School of Architecture, Design and Planning. His research is at the intersection of interaction design, community engagement, urbanism, and smart cities. Joel has worked on a variety of transdisciplinary research projects that incorporate participatory methods to design, develop and deploy interactive systems that enhance community engagement and contribute to collaborative city-making. Joel has authored and co-authored in numerous publications and books. Most notably he was the lead author of Media Architecture Compendium Vol.2 – Concepts, Methods, Practice, which draws on academic research and global studies to present an evolutionary account of concepts, methods, and practice for bringing media architecture thinking into projects.

Tram Thi Minh Tran is a doctoral candidate in the Urban Interfaces Lab at the University of Sydney's School of Architecture, Design, and Planning. Her research explores the applications and implications of emerging technologies, with a particular focus on AR/VR and autonomous mobility.

Julie R. Williamson is a senior lecturer at the University of Glasgow. Her work explores how we experience immersive realities together using social signal processing to analyse and model behaviour in social XR. She is also involved in digital accessibility in her roles on the ACM Digital Library Board and CHI Steering Committee.

Youngho Lee is a Professor in the department of computer engineering at Mokpo National University, South Korea. His research interests include virtual and augmented reality, culture technology, and human-computer interfaces. Currently, he is focused on designing, implementing, and evaluating training systems for nurses with virtual and augmented reality technology.

Woontack Woo is the Director of the Post-Metaverse Research Center and a Professor at the Graduate School of Metaverse at KAIST in Daejeon, Korea. In 2001, he coined the term "ubiquitous virtual reality" and has since published over 600 papers/patents on topics such as Augmented Reality, Augmented Humans, and

related subjects. His current research interest lies in exploring how to Augment Society with the Metaverse.

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